SESSION 5: OTHER METHODS OF SHARE VALUATION

Learning Objectives

At the end of this session, you should be able to:

a) determine the value of a share of a business using the super-profit method
b) apply the capital asset pricing model [CAPM] in share valuation.
c) calculate the value of a share of a business using the dual capitalization method.

The Super Profit Method

By this method, a fair rate of return is applied to the net tangible assets and the result is compared with the expected profits. Any excess of profit (referred to as super profit) is used to calculate goodwill. The goodwill calculated at a specified number of years' super-profit is added to the value of target company's net assets to arrive at a value for the business,

Illustration 1

A Ltd has net tangible assets of GH¢48,000 and earnings of GH¢16,000. B Ltd wants to take over A Ltd and considers that a fair returns for this type of industry is 30% and decided to value A taking Goodwill at 4 years' purchase of super profit. Determine the value of A Ltd

Solution 1

Actual Earnings	GH¢16,000
Expected Earnings (30% of ¢48,000)	GH¢14,400

Super Profit	GH¢ 1,600

Goodwill = 4 x GH¢1,600 = GH¢6,400

Value of A Ltd = Net Assets + Goodwill = GH¢48,000 + GH¢6,400 = GH¢54,400

The main limitations of this method are as follows:

- The expected rate of return is subjectively or arbitrary chosen without any scientific precision;
- The number of years purchase of super profit is arbitrary

The Capital Asset Pricing Model (CAPM)

The CAPM may be used to value shares, particularly, when pricing shares for a stock market listing. The CAPM may be used to establish a required equity yield.

Illustration 2

A Ltd is planning to obtain a stock market listing by offering 40% of its existing shares to the public. No new shares will be issued. Its most recent summarized results are as follows:

Turnover	GH¢240,000m
Earnings	GH¢3,000m
Number of shares in issue	6 million

The company is lowly geared and has a dividend policy of 50% pay-out rate. And the retention policy is expected to achieve 10% dividend growth each year.

Summarized details of two listed companies in the same industry as A Ltd are as follows:

	B Ltd	C Ltd
Gearing (Total debt/ Equity)	45%	10%
Equity Beta	1.60	1.10

The current Treasury bill yield is 20% per annum. The average market return is estimated to be 25%. The shares will be offered to the public at a price 15% lower than the estimated market valuation in order to increase the prospects of success for the public issue.
What will be the issue price?

Solution 2
Using the CAPM, we begin by deciding on suitable beta value of A Ltd equity. We shall assume that since A Ltd's gearing is close to C's, a beta of 1.10 is appropriate.

The cost of A Ltd equity = Rf + B(Re – Rf)
=20% + 1.10(25% -20%)
=25.5%

This can now be used in the dividend growth model. The dividend this year is 50% of GH¢3,000 m = GH¢1,500m
The total value of A Ltd equity is MV = $\frac{D0\ (1+g)}{r-g}$

= $\frac{GH¢1,500\ m(1.1)}{0.255-0.10}$

= $\frac{GH¢1,650\ m}{0.155}$

= GH¢10,645 m

Value per share = GH¢10,645/6million = GH¢1,774
Since the shares that are offered to the public will be offered at a value 15 % below the valuation , the share price for the market launch will be 85% of GH¢1,774 =¢1,508, approximated to about GH¢1,500

Berliner Method
This method tries to combine earnings potential of a business entity and the value of its net assets to arrive at the business valuation

- The earnings potential is arrived at by capitalizing expected future profits using a standard /suitable /acceptable /expected rate of return
- The net assets valuation is based on going concern values
- The simple mean of the two valuations is taken to be the value of the entity.

This method tries to establish a compromise between the net asset method and the earnings method in arriving at an acceptable share or business purchase price, it however lacks theoretical base.

Dual Capitalisation Method

This method is used to determine acceptable yields for both tangible and intangible assets. An acceptable rate of return is applied on value of tangible assets. The figure ascertained represents return on tangible assets. The difference between this figure and the earnings of the business represents returns on intangible assets. The earning attributable to intangible assets is capitalized at the appropriate rate and the result represents the 'unseen' value or inherent goodwill of the business. The unseen value/ inherent goodwill value is then added to the value of tangible asset previously calculated to arrive at total value of the entity

Illustration 3

ABC Ltd's average annual earning is about GH¢70,000 and this approximates its maintainable earnings. The total tangible assets of the business values GH¢300,000 and total liabilities is GH¢50,000. The required rate of return on net tangible assets is 20%. As a result of higher risk associated with investment in intangible assets, required rate of return thereon is pegged at 32%.

Ascertain the value of ABC Ltd using the dual capitalization method

Solution 3

	GH¢
Annual Earning	70,000
Expected return on Net Tangible assets	
20% of (GH¢300,000 - GH¢50,000)	50,000

Returns attributable to intangible assets	20,000
	=====

Capitalized value of Intangible assets = GH¢20,000/ 0.32 = GH¢62,500

Thus, Value of ABC = value of net tangible assets + value of intangible assets

= GH¢250,000 + GH¢62,500 = GH¢312,500

SUMMARY OF METHODS OF SHARE/BUSINESS VALUATION

METHOD	SHARE VALUE	REQUIRED DATA	ADVANTAGES	PROBLEMS	SUITABLE FOR
Earnings Yield	Earnings x PE Ratio Earnings/Earnings Yield	Future Earnings PE Ratio Expected	Independent of mgt's dividend policy.	Forecasting earnings Deciding on PE ratio Subjectivity of earnings Effect of inflation on earnings	Quoted Companies Shareholdings in private companies of a size which enables the holder to influence dividend policy
Dividend Yield	Dividend (gross) X 100 Dividend yield % (gross)	Future dividends Dividend yield expected	Cash paid out is the figure Objective	Forecasting dividends Deciding on the Dividend Yield	Quoted Companies Small shareholdings (up to about 20%) in

	$\frac{D_0(1+g)}{R-g}$			Mgt's dividend policy.	companies with stable dividend policy
Assets Basis	Net Asset Value	Asset & Liability valuations	Useful check on other methods	Assigning values to individual assets Limited Applicability	Going concern basis – as a check on earnings yield method Asset-stripping - useful information if assets are valued at disposal value

Illustration Question 4

In 2025 the shareholders of Boafo Ltd decided to dispose of their shares in the company. The summarized financial statements of Boafo for the year 2025 are given below:

Statement of Financial Position as at 31 December 2025

	GH¢m
Patents	10,000
Tangible non-current assets	146,000
Current assets	53,500
Current Liabilities	(32,000)
Long term loans	(87,500)

	90,000
	======
Stated Capital	50,000
Income Surplus	40,000

	90,000
	======

Statement of comprehensive income for the year ended 31 December 2025

	GH¢m
Profit before Tax	62,000
Tax	20,000
	42,000
Proposed Dividend	8,000
Retained Profit	34,000
	======

Additional Information

- The stated capital of Boafo Ltd is made up of 20,000,000 ordinary shares of no par value. The company is not listed.
- The fair value of tangible non-current assets of Boafo Ltd was GH¢290 billion. Current assets include obsolete inventory of GH¢16.5 billion.

- The patents represent a license to produce and sell a special product. This product is expected to generate a pre-tax profit of GH¢2 billion per annum over the next five years.
- The discounted present value of future cash payments in respect of the long term loan is GH¢122 billion.
- Profits after tax of Boafo Ltd over the past 4 years were as follows:

Year	GH¢m
2021	28,000
2022	30,000
2023	16,000
2024	36,000

- The discount rate of Boafo Ltd is 10% per annum
- Happy Ltd, a major competitor of Boafo Ltd, is listed with a P/E ratio of 9 and dividend yield of 5.2%

Required

i) Describe any three (3) suitable situations that may lead to the valuation of shares

ii) Using three (3) methods of valuation, advise the directors of Boafo Ltd on the values to be placed on the ordinary shares

Solution 4

a) Situations that may lead to valuation of shares include the following:

- The company wishes to go public and get listed on the exchange and therefore must ascertain an issue price for the share issue;
- There is a scheme of merger and each company's shares must be assessed;
- Shares are being sold;
- Shares are being pledged as collateral for a loan

b) Valuation of Shares of Boafo Ltd

The value to be placed on equity shares of Boafo Ltd depends on the method of valuation used. The following methods of valuation can be applied;

i) Earnings Yield / Price- Earnings Method:

Value per share = EPS X P/E ratio

Earnings may be the most recent earnings, the simple average of the past five years ' earnings, or weighted average of the past five years ' earnings. In the absence of a specific instruction, any of the earnings figure may be appropriate for the valuation. Granting that the most recent years earnings is chosen,

EPS = GH¢42,000M / 20 million shares = GH¢2,100

P/E Ratio = The P/E of a similar company adjust for higher risk as a result of lack of marketability = 9 **adjusted to say 8**

Value per share = GH¢2,100 x 8 <u>= GH¢16,800</u>

ii) Dividend yield Method

Value per share = Dividend per share / Dividend yield

Dividend per share = GH¢8,000 m / 20 m shares = GH¢400

Dividend yield = That of a similar quoted entity, adjusted to give recognition to lack of marketability = 5.2% (**adjusted to say 6%)**

Value per share = GH¢400 / 0.06 = GH**¢6,667**

III) Net Asset basis

Assets	GH¢'m
Tangible non-current assets	290,000
Current Assets	37,000
Patents rights (3.790 x 2,000)	7,580
Current liabilities	(32,000)
Long term loans	(122,000)
Net Assets	180,500

value per share = GH¢180,500 m / 20m = GH¢9,025

Illustration 5

ABC Ltd is a private company, whose owners are also the directors. The directors have decided to sell their business and have began a search for organisations interested in the purchase. They have asked for your assessment of the price per ordinary share a purchaser might be expected to offer.

Relevant information is as follows:

Statement of Financial Position as at 31 December 2024

`	GH¢'M	GH¢'M
Land and Buildings		800
Plant and equipment		450
Motor vehicles		55
Patents		2
		1,307
Current assets		
Inventories	250	
Receivables	125	
Cash	8	
	383	
Current Liabilities		
Payables	180	
Taxation	50	
	230	
Net Current assets		153
Loan secured on property		(400)
		1,060

Stated capital	(300,000 ordinary shares)	300
Income Surplus		760
		1,060

The profits after tax and interest but before dividends over the last five years [including 2024] have been as follows

	GH¢m
2020	80
2021	78
2022	87
2023	95
2024	100

The annual dividend has been GH¢45 million (gross) for the last 6 years
The company's five year plan forecasts an after tax profit of GH¢100 million for the next 12 months with an increase of 4% a year over each of the next 4 years.
As part of their preparations to sell the company, the directors of ABC Ltd have had the non-current assets re-valued by an independent expert with the following results:

	GH¢m
Land and buildings	1,075
Plant and machinery	480
Motor vehicles	45

The dividend yields and the P/E ratios of three quoted companies in the same industry as ABC over the last three years have been as follows

	A LTD		B LTD		C LTD	
	D/y %	P/E	D/Y %	P/E	D/Y %	P/E
2024	12.0	8.5	11.0	9.0	13.0	10.0
2023	12.0	8.0	10.6	8.5	12.6	9.5
2022	12.0	8.5	9.3	8.0	12.4	9.0
Average	12.0	8.33	10.3	8.5	12.7	9.5

Large companies in the industry apply an after-tax cost of capital of about 18% to acquisition proposals when the investment is not backed by tangible assets, as opposed to a rate of only 14 % on the net tangible assets.

Your assessment of the net cash flows which will accrue to a purchasing company. Allowing for taxation and the capital expenditure required after the acquisition to achieve the company's target five-year plan, is as follows:

Year	Cash flow(¢m)
1	120
2	120

3	140
4	70
5	120

Required
Use the information provided to suggest six valuations which prospective purchasers might make

Solution 5
Earning Based Valuations

1) **Earnings Yield**
Using this method, the value per share is calculated as Earnings per share (EPS) X Price Earning (P/E) ratio

- Earnings can be taken as the average of the past years, the most recent profit or the future earnings. Assuming that the most recent profit is taken the EPS is ¢100m / 0.3 m shares =

- P/E is taken as the average of the recent P/E ratios of the three similar listed entities. I.e (8.5+9.0+10.0) / 3 = 9.67
  Discounted to give recognition to lack of marketability; say 8

- Value per share =GH¢333.33 x 8 = GH**¢2,667**

2) **Dividend Yield**
Using this method**,** the value per share is calculated as Dividend / Dividend yield

- Dividend per share may be taken as GH¢45m /0.3m shares GH¢150

- Dividend yield may be taken to be the average of the three similar listed entities i.e (12% + 11% +13%) / 3 = 12%
  Discounted to give recognition to lack of marketability say 15% (note that the higher the risk, the higher the expected returns)

  - Value per share = GH¢150 / .15 = GH¢1,000

3) Dividend growth method

Using this method, the value per share is calculated as D (1+g) / (r – g)

= 150(1+.04) / (.15 -.04)

= 156 / 0.11

= GH¢1,418

4 Discounted Cash flow method

Year	Cash flow (GH¢m)	Disc Factor	Present Value (GH¢m)
1	120	0.877	105.24
2	120	0.769	92.28
3	140	0.675	94.50
4	70	0.592	41.44
5	120	0.519	62.28

			395.74
			======

Value per share = GH¢395.74m / 0.3m shares = GH¢1,319

Assets Based Valuation

5 **Replacement Cost Based Valuation**

The value per share is calculated as the Net Assets (based on Replacement Cost basis) divided by number of equity shares

		GH¢'M
Land and Buildings		1,075
Plant and Machinery		480
Motor Vehicles		45
Patents		2
Inventories		250
Receivables		125
Cash		8

		1,985
Liabilities	(180+50+400)	(630)

Net Assets		1,355
		=====

Value per share = GH¢1,355 m / 0.3 m shares = GH¢4,517

6) Book Value based valuation

Value per share is net book value/number of shares= GH¢1,060m /0.3m shares = GH¢3,533